



# HarvestWise

Voice-first autonomous restocking for India's street merchants.

WhatsApp / Soundbox



Sarvam Indic Voice AI



Cognee Knowledge Graph



n8n Orchestrator



Mandi Wholesaler

Lakshmi sells vegetables in Bengaluru. Paytm made her a **merchant** — nothing made her a **restocker**. HarvestWise listens, predicts spoilage, recalculates her order and dispatches it to her mandi wholesaler — **without asking her to type a single word.**

# Paytm made them merchants. Nobody made them restockers.

India's largest merchant network is also its least-served growth market. Payments are digital, but the highest-stakes daily decision — what to buy at 4:30 AM — still runs on guesswork.

## 30M<sub>+</sub>

### MERCHANTS IN PAYTM'S ECOSYSTEM

The long tail: kiranas, carts, stalls — the backbone of India's daily commerce.

## 1.57<sub>Cr</sub>

### MERCHANT DEVICE SUBSCRIPTIONS (SOUNDBOX / POS)

A proven billing rail merchants already pay on — and the anchor for cross-sell.

## 8<sub>Cr</sub>

### MONTHLY TRANSACTING USERS

Demand-side scale that makes every merchant insight immediately actionable.

### The locked layer of value

Paytm sees the transactions — but the merchant's most consequential decision, **what to stock tomorrow at what quantity**, happens blind at the mandi gate. Paytm's own AI (ARMS) optimizes merchant lifecycle *internally*; the merchant's own P&L still runs on intuition. That missing merchant-facing layer is where the next phase of Paytm's growth lives.

**Why now:** Indic voice AI (Sarvam), agent orchestration (n8n) and knowledge-graph memory (Cognee) matured together for the first time, so a copilot for a non-English, non-dashboard user is finally buildable.

# Her inventory rots. Her margin goes with it.

"I'm at Kalasipalya mandi by 4:30 AM with ₹3,500 in my pocket. I buy tomatoes, greens and beans on one heuristic: what sold yesterday. At 3 PM an unpredicted thunderstorm empties the street. By 9 PM I'm dumping ₹400 of rotting greens in the garbage. I earn on Paytm. I restock on guesswork."

— LAKSHMI · VEGETABLE CART · BASAVANAGUDI, BENGALURU



Modeled across Bengaluru's 12,000+ vegetable vendors: ~₹175 Cr/yr in spoilage losses (₹400 × 12,000 × 365) — losses like these push vendors into informal high-interest debt (*meter baddi*). Baseline to be validated in pilot.



## 25–35% daily spoilage

Perishable stock bought at 4:30 AM on intuition, sold across a 12-hour day. Evening leftovers go out at distress prices — what doesn't sell is a 100% cash loss.



## 4:30 AM <—> 3 PM disconnect

Procurement decisions ignore the signals that actually move demand: afternoon rain, micro-festivals, local UPI purchase velocity, mandi price swings — all available, none connected.



## Tools not built for her

Dashboards need literacy, dry hands and downtime she doesn't have. Chatbots only talk. Wet hands, continuous customers, vernacular-only — the interface must be speech.

₹3,500 / day

AVG. WORKING CAPITAL



~30%

UNSOLD AT CLOSE



evening salvage

DISTRESS DISCOUNTING



₹400 / day

NET LOSS · ~₹12,000/MO

Up to 30% of household take-home erased  
· illustrative model, spoilage band 25–35%

# A restocking loop that speaks, acts & remembers

## 1 • OBSERVE



**Watches her world** — hourly UPI sales velocity (Paytm signals, with consent), IMD rain probability, festival calendar, mandi rates, and **perishable decay tables** per commodity.

## 2 • PREDICT



**Forecasts spoilage & demand** in her words: *"Rain at 3 PM — coriander shelf-life drops to 4 hours; your rainy-day coriander sales fall 40%."*

## 3 • ACT



**Recalculates and dispatches** — resizes her order (10 → 6 bundles), confirms with her by voice, sends the revised buy-order to her mandi wholesaler on WhatsApp.

## 4 • LEARN



**Remembers her business** — which recalcs worked, which didn't, and why. Next week's order starts from this week's evidence, not zero.

### No new app. No login. No literacy barrier.

The interface is **voice on WhatsApp** — plus the morning briefing **played on her Paytm Soundbox**. The Soundbox already sits on her cart, already speaks her language. We upgrade it from payment confirmation to operational assistant.

### Not another chatbot

Chatbots **advise** ("stock less"). HarvestWise **finishes the job** — it recalculates the order and dispatches it to the wholesaler, with a human voice-approval gate before anything that touches money.

### One loop, every morning: Observe → Predict → Act → Learn.

Spoilage is prevented before it's bought, orders go out before the mandi gate opens, and the same evening the outcome is measured — her take-home improves, and so does Paytm's GMV.

# A day with HarvestWise — in Kannada

The product story — the finale demo compresses it to one voice command → one explained recalculation → one dispatched order → one remembered outcome.

6:30

## AM · BRIEFING

Namaskara Lakshmi! Today: 80% chance of rain at 3 PM. Order fewer tomatoes; leafy greens will sell out by evening. Best mandi rate: ₹18/kg at Kalasipalya.

Auto-sent by the 6:30 AM n8n workflow — voice on WhatsApp **and played on her Soundbox**. 30 seconds, zero taps.

9:14

## AM · SHE ORDERS

ಟೋಮ್ಯಾಟೋ 3 ಬಾಕ್ಸ್, ಕೊತ್ತಂಬರಿ 10 ಕಟ್ಟು.  
("3 crates of tomatoes, 10 bundles of coriander.")

Order noted. One flag: rain at 3 PM — coriander lasts only 4 hours in humidity, and your rainy-day coriander sales drop ~40%.

Sarvam ASR → intent + entities → Cognition raises the spoilage flag against her history.

11:02

## AM · IT RECALCULATES

Revised: 6 bundles of coriander (not 10), 3 crates of tomatoes. Wholesaler Ravi anna — ₹546. Approve?

ಹೌದು, ಕಳುಹಿಸು. ("Yes, send it.")

**Dispatched:** buy-order sent to the wholesaler, stock sheet updated — n8n gate: nothing moves without her "yes".

8:30

## PM · IT LEARNS

Today: sold out by 7:15 PM, zero coriander waste. You kept ₹1,940 you'd have discounted away. This week: +₹4,100 vs last week.

Logged to her Cognition memory graph: **"Rain recalc — worked, ₹1,940 kept."** Next storm, she doesn't have to think.

# Built on the organizers' own stack — one job each

## Experience

WHERE THE MERCHANT LIVES

### WhatsApp Business API

voice notes, in & out

### Paytm Soundbox

morning audio briefing

### Business with Paytm

optional dashboard view

No new app, no login, no typing. Voice is the entire UI; the Soundbox is the loudspeaker.

## Intelligence

SARVAM AI · INDIC SPEECH LAYER

### Sarvam Speech-to-Text

ASR · 22+ Indic languages

### Sarvam LLM

intent + entities + reasoning

### Sarvam Text-to-Speech

natural voice · 11 languages

Indic-native understanding — not an English product with translation bolted on.

## Decision Core

COGNEE · CAUSAL KNOWLEDGE GRAPHS

### Merchant memory graph

her history · what worked

### Commodity decay tables

shelf-life × humidity × rain

### Cold-start patterns

anonymized cross-merchant

Shelf-life × rain × her UPI velocity — the causal layer behind every recalc.

## Orchestration

N8N · EVENT-DRIVEN ACTIONS

### n8n 6:30 AM workflow

briefing · WhatsApp + Soundbox

### n8n order dispatcher

revised buy-order → wholesaler

### n8n approval gates

no money action without "yes"

Event-driven dispatch to mandi wholesalers; escalation & fallback built in.

## Data

CONSENT-FIRST SIGNALS

### Paytm merchant signals

hourly UPI velocity · consent

### IMD weather API

afternoon rain probability

### Mandi rate feeds

public daily price feeds

Demo runs on Paytm-transaction-shaped seeded data (labeled as such); production stays consent-first inside Paytm's rails.

■ **GUARDRAILS ARE THE PRODUCT** DPDP-aligned · federated learning (patterns shared, raw data never) · human approval on every money action · every recalc explained & undoable · full audit trail

# Advice is cheap. Closed loops are rare.

Every alternative stops at a screen or a sentence. HarvestWise speaks her language, recalculates her order, dispatches it, and remembers her business.

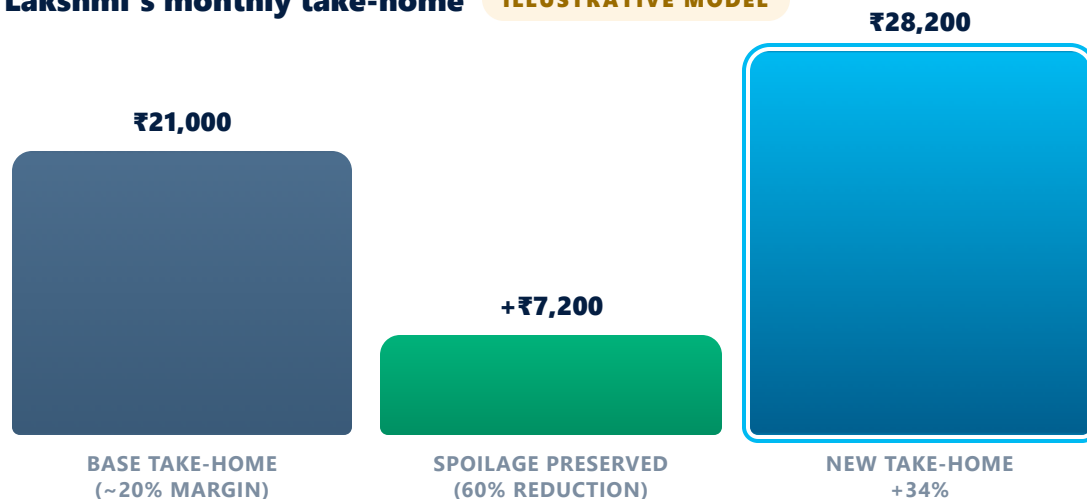
| Evaluation            | Traditional merchant tools                                       | Generic AI chatbots                           | HarvestWise                                                                                         |
|-----------------------|------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Input modality        | English forms & dashboards — needs literacy, dry hands, downtime | Text-only chat — assumes she can type         | <b>Zero-touch Indic voice notes</b> on WhatsApp (Sarvam) — she talks; that's the whole UI           |
| Decision intelligence | Static inventory tallying — backwards-looking                    | Generic public advice — no context, no memory | <b>Causal knowledge graph:</b> shelf-life × rain × her hourly UPI velocity (Cognee)                 |
| Workflow completion   | Manual execution — she still does the work                       | Advises only — "stock less", then stops       | <b>Closed-loop procurement:</b> recalculated buy-order dispatched to her mandi wholesaler (n8n)     |
| Hardware tie-in       | None                                                             | None                                          | <b>Morning audio briefing on Paytm Soundbox</b> — payment speaker upgraded to operational assistant |

**MOAT** Per-merchant memory graph (Cognee) × vernacular closed loop (Sarvam + n8n) × Soundbox/Paytm-native distribution — a compounding data advantage every single day.

# Waste is her biggest expense. We give 60% of it back.

## Lakshmi's monthly take-home

ILLUSTRATIVE MODEL



Assumes ₹3,500 avg. daily working capital × 30 selling days, ~20% gross margin, ₹400/day net spoilage loss after evening salvage. The 60% spoilage-reduction target (₹240/day ≈ ₹7,200/mo) is the pilot metric we commit to validating — not a claim.

## What Paytm wins

- Soundbox upgraded:** from passive payment-confirmation speaker to **active operational assistant** — a deeper, stickier device relationship.
- GMV up:** better-stocked merchants capture demand they currently miss — target **+8% GMV per merchant**.
- Underwriting data moat:** proprietary inventory & procurement telemetry de-risks **Paytm Merchant Loans** — replacing "meter baddi" with formal credit.
- Mission delivered:** financial inclusion that shows up as a healthier family P&L — the story of Build for India.

₹7,200

PRESERVED / VENDOR / MO\*

₹105 Cr+

WORKING CAPITAL / YR · 12K VENDORS\*

₹175 Cr+

CURRENT ANNUAL SPOILAGE LOSS (CITY-WIDE EST.)

\*Model: 12,000 vendors × ₹240/day × 365 — directional; validated in pilot. Impact framed as money preserved, not vanity metrics.

# Billed where she already pays. Free where she starts.

## Seed

₹0 / forever

- **Daily demand alerts** — weather + spoilage risk, one voice note + Soundbox briefing
- Yesterday's sales recap
- Builds the daily habit — **zero CAC** via Paytm for Business & Soundbox base

## SETTLEMENT-BILLED

## Pro

₹149 / month

- **Autonomous restocking:** order recalculation + dispatch to her wholesaler
- Price-drop supplier alerts · end-of-day **P&L voice audit**
- Proposed billing: **settlement-linked micro-deduction** — subject to Paytm integration approval; **simulated in our prototype**. No invoice, no friction.

## Platform Revenues

**Take-rate** on value created

- **Supplier marketplace:** % on executed mandi orders
- **Credit distribution:** procurement telemetry de-risks Paytm Merchant Loans
- **Anonymized demand curves** (DPDP-compliant) for supply-chain aggregators & mandi logistics planning

## Why merchants buy

₹149/mo vs ₹7,200/mo preserved — pays for itself in one saved evening of stock.

## Why it compounds

Every day improves her memory graph — advice quality compounds, churn falls.

## Unit economics

Paytm-native distribution keeps acquisition at zero cost; SaaS margins run 80%+; and the data flywheel feeds straight into lending.

## Why Paytm wins

The deepest merchant lock-in since the Soundbox: the device confirmed payments; we grow them.

**THE WEDGE → THE PLATFORM** Start: perishable carts & stalls (highest pain, clearest signal) → kirana fresh categories → 30M+ merchants as the daily restocking layer of the Paytm ecosystem.

[NEXT](#)[ROADMAP](#) · [TEAM](#) · [ASK](#)

# From this deck to a pilot

## NOW — FINALE

### MVP on partner stack

Synthetic 90-day sales data; 3 n8n workflows (briefing, order recalc + dispatch, day-wrap learning); Sarvam voice loop; Cognee memory. Live demo, one phone.

## MONTHS 1-3

### 500-merchant pilot

KR Market, Russell Market & Madivala with vendor associations + Paytm's merchant network. Validate the 60% spoilage-reduction metric.

## MONTHS 4-6

### Paytm-native integration

API partnership with Paytm for Business; Soundbox briefing rail; mandi logistics channels wired into n8n dispatch.

## BEYOND

### The restocking layer

Perishables → kirana fresh → all categories. Procurement telemetry → merchant credit. Every merchant, a copilot.

"Every morning at 4:30, India's merchants guess. From tomorrow, they guess **with a copilot.**"

## TEAM

Sairam — Venkata Narayana G V · Malaravan P

## BUILT WITH

Sarvam AI · n8n · Cognee

## ASK

A 500-merchant pilot with Paytm's network

Solve with AI. Build for India.